

Introduction



This book concentrates on the math tackled in schools between the ages of 9 and 16. But it does so in a gripping, engaging, and visual way. Its purpose is to teach math by stealth. It presents mathematical ideas, techniques, and procedures so that they are immediately absorbed and understood. Every spread in the book is written and presented so that the reader will exclaim, "Ah ha—now I understand!" Students can use it on their own; equally, it helps parents understand and remember the subject and thus help their children. If parents too gain something in the process, then so much the better.

At the start of the new millennium I had the privilege of being the director of the United Kingdom's **Maths Year 2000**, a celebration of math and an international effort to highlight and boost awareness of the subject. It was supported by the British government and Carol Vorderman was also involved. Carol championed math across the British media, and is well known for her astonishingly agile ways of manipulating and working with numbers—almost as if they were her personal friends. My working, domestic, and sleeping hours are devoted to math—finding out how various subtle patterns based on counting items in sophisticated structures work and how they hang together. What united us was a shared passion for math and the contribution it makes to all our lives—economic, cultural, and practical.

How is it that in a world ever more dominated by numbers, math—the subtle art that teases out the patterns, the harmonies, and the textures that make up the